Multimodal Retrieval-Augmented Generation for Chart and Document Understanding

Abstract

Multimodal retrieval-augmented generation for chart and document understanding has emerged at the intersection of document AI, chart understanding, multimodal retrieval, and large multimodal language models. The central problem is no longer merely to read a single page or answer a question about a single chart. Instead, modern systems must identify, retrieve, align, and reason over heterogeneous evidence scattered across pages, regions, tables, figures, charts, and external corpora, often under severe context-length, latency, and faithfulness constraints. This survey develops a unified perspective on that problem space. We argue that the field is best understood as a four-stage pipeline—representation, indexing, retrieval, and grounded generation—whose design choices differ according to document modality, evidence granularity, and reasoning style. We review enabling foundations in layout-aware document modeling, OCR-free document understanding, chart-specific pretraining, multivector visual retrieval, hierarchical and region-level evidence selection, graph-augmented retrieval, and agentic reasoning. We further synthesize the rapidly growing benchmark ecosystem spanning single-page document QA, multi-page and multi-document QA, scientific-paper understanding, and chart-centered MRAG evaluation. Across these literatures, a recurring pattern emerges: systems perform best when they preserve visual fidelity, retrieve at multiple granularities, and maintain explicit grounding between retrieved evidence and generated answers, but they remain brittle under distribution shift, visually similar templates, multimodal conflict, and adversarial corruption. We conclude by identifying open problems in unified multimodal indexing, citation-grade grounding, efficiency for long and enterprise-scale corpora, chart-specific reasoning under real-world variation, and benchmark design that captures evidence sufficiency rather than answer overlap alone. [1-19,20-48,52-66,74-101,109-164] [1]

Introduction

Retrieval-augmented generation was originally developed to couple parametric generation with non-parametric memory, improving knowledge access, reducing hallucination, and supporting domain adaptation beyond the model's training cutoff [1,3-6]. In multimodal settings, however, the "memory" is not a homogeneous text collection. It may be a scanned page, a born-digital PDF page, an OCR token stream with coordinates, a chart image, a derived table, a slide deck, or a scientific paper whose answer requires joint reading of prose, equations, tables, and figures [2,7-15]. For chart and document understanding, therefore, MRAG is not simply RAG plus images; it is a family of systems that must decide **what** to represent, **how** to index it, **which** evidence unit to retrieve, and **how** to condition a generator on evidence that remains spatially and semantically grounded. [1-15,17-19] [2]

This problem setting is motivated by two converging developments. First, document AI has moved from OCR-dependent token-layout models such as LayoutLM, LayoutLMv2, LayoutXLM, LayoutLMv3, DocFormer, and TILT toward OCR-free or weak-OCR multimodal models such as Donut, Pix2Struct, Nougat, mPLUG-DocOwl, and related document MLLMs [20-40]. Second, chart understanding has evolved from synthetic chart QA benchmarks such as FigureQA and DVQA toward real-world benchmarks and chart-specialist models such as ChartQA, OpenCQA, DePlot, MatCha, UniChart, ChartAssistant, ChartInstruct, TinyChart, ChartX, and ChartBench [67-89]. MRAG inherits the representational burdens of both lines of work: it must preserve spatial and visual detail as document models do, while also supporting the structured numerical and relational reasoning demanded by chart understanding [74-101]. [20-40,67-89] [3]

A second source of difficulty is scale. Many contemporary benchmarks now evaluate multi-page, multi-document, or open-domain document understanding over corpora that are too large to fit into a single multimodal context window [52-66]. Benchmarks such as DUDE, MMLongBench-Doc, M-LongDoc, LongDocURL, M3DocRAG, VisDoM, and VDocRAG make it increasingly clear that direct long-context processing alone is not a sufficient answer; systems must retrieve evidence selectively, fuse modalities economically, and maintain citation-grade traceability of retrieved support [56-66,111-121]. In chart-centric MRAG, the situation is even sharper: recent work shows that models often exhibit text-over-visual bias, underperform on chart-only or text-chart questions, and degrade markedly when visual evidence must be retrieved rather than supplied as gold context [74-97,115,116,164]. [56-66,111-121,164] [4]

The goal of this survey is therefore twofold. First, it organizes multimodal RAG for chart and document understanding into a coherent taxonomy that spans representation, retrieval, generation, and evaluation. Second, it treats chart understanding not as a side topic but as a stress test for document-centric MRAG: charts expose failures in low-level perception, structured extraction, numerical reasoning, and evidence grounding more starkly than many text-rich document tasks. Throughout, we emphasize that the most relevant literature includes not only end-to-end MRAG systems, but also upstream work on document and chart encoders, visual document retrieval, and evidence-grounded evaluation, because those components define what MRAG systems can retrieve and how credibly they can answer. [9,17-19,20-48,67-101,109-164] [1]

Methods

Problem Formulation and Taxonomy

A useful formalization views multimodal RAG for chart and document understanding as a quadruple (E, I, R, G): an evidence representation space E, an index I built over evidence units, a retriever R that maps a query to a ranked multimodal evidence set, and a generator G that conditions on that evidence to produce an answer, explanation, citation, or structured output. In conventional text RAG, evidence units are usually passages. In document and chart MRAG, the evidence unit can instead be an OCR token span, a layout block, a page screenshot, a chart region, a chart-derived table, a multi-page cluster, or a node in a graph over document elements [1-15,20-32,77-79,109-147]. This means that the core architectural question is not only retrieval quality, but also **evidence granularity**: coarse units improve efficiency but often suppress fine visual cues; fine units improve precision but raise indexing cost and fusion complexity.

A second taxonomic axis is **modality of retrieval**. At one end are parse-first systems, which convert documents into text, layout tokens, tables, chart tables, or structured markup, then use largely textual or structured retrieval [20-31,33,35-43,77-79]. At the other end are vision-first systems, which index screenshots or vision-language embeddings directly, avoiding early information loss from OCR or chart derendering [34,39,40,57-63,109-114,120-132]. Between them lie hybrid systems that index multiple synchronized views of the same source document: OCR text, page image, chart table, region crops, and graph-structured metadata. These systems attempt to exploit complementary strengths: parse-first pipelines provide lexical precision and symbolic manipulation, while vision-first pipelines preserve typography, layout, color, legends, marks, and small visual cues that are frequently decisive in charts and visually rich documents [77-90,109-145]. [20-40,57-63,77-90,109-145] 5

A third axis is **retrieval topology**. Single-stage retrieval is common in early systems, but recent work increasingly adopts hierarchical, coarse-to-fine, or iterative designs. CREAM retrieves candidate pages and then refines answer generation [113]; M3DocRAG, VisDoM, and VDocRAG extend retrieval across multiple pages and documents [112,120,121]; MMRAG-DocQA, RegionRAG, HKRAG, and related methods perform page-level or document-level pruning before region- or block-level evidence selection [123,136,137]. Graph-augmented systems such as RECON, MKG-RAG, and LILaC add relational structure among pages, sections, or multimodal entities [145-147]. Agentic variants such as ViDoRAG, HM-RAG, HEAR, MDocAgent, and MoLoRAG make retrieval itself a deliberative process, often alternating query rewriting, evidence inspection, and refined retrieval steps [139-145]. The field's most consistent trend is therefore away from monolithic "retrieve once, answer once" pipelines and toward **multi-step evidence construction**. [112,113,120-147] 6

Representation and Retrieval Architectures

The representation layer has inherited two major traditions from document AI. The first is **layout-aware token modeling**, where text tokens, bounding boxes, and sometimes region-aligned visual features are fused into multimodal encoders. LayoutLM, LayoutLMv2, LayoutXLM, LayoutLMv3, DocFormer, BROS, LiLT, StrucTexT, ERNIE-Layout, UniDoc, and TILT all instantiate variations of that idea [20-31]. These models remain highly relevant to MRAG because parse-first retrievers depend on representations that can preserve reading order, region boundaries, and local document structure. For forms, receipts, invoices, and moderately structured PDFs, token-layout encoders still offer a favorable trade-off between precision and efficiency, particularly when lexical match matters or when downstream retrieval must support symbolic filtering [20-32,49-56].

The second tradition is **OCR-free visual sequence modeling**. Donut, Pix2Struct, Nougat, DocParser, UReader, mPLUG-DocOwl, DocO-Pilot, and related models treat screenshots or page images as the primary input and learn either direct structured generation or multimodal reasoning over visual tokens [33-48]. For MRAG, these models expand the design space in two ways. First, they support vision-first indexing, as exemplified by screenshot embeddings and ColPali-style retrieval [109,110]. Second, they reduce early commitment to parser outputs, which is valuable in heterogeneous corpora containing diagrams, iconography, small print, dense tables, and charts whose semantics may be degraded by OCR-only conversion. The trade-off is storage and retrieval cost: page screenshots and patch embeddings consume far more memory than text embeddings, motivating pruning and hybrid-vector strategies such as DocPruner, storage-efficient patch selection, and hybrid single-/multi-vector retrieval [109,128-132]. [33-48,109-132] 7

Chart understanding introduces a third representational family: **chart-specialized structural modeling**. Early benchmarks such as FigureQA, DVQA, LEAF-QA, and PlotQA largely exposed perception errors and vocabulary limitations [67-70]. Later work split into three methodological streams. One stream focuses on **direct chart QA** using chart images and question text [74,79-90]. A second emphasizes **chart derendering**, converting a chart into an intermediate table or code representation; DePlot, MatCha, ChartOCR, and related work show that chart-to-table conversion can substantially simplify downstream reasoning, especially arithmetic and comparative reasoning [73,77,78]. A third stream emphasizes **instruction tuning**

and tool use, where general or chart-specialized MLLMs are exposed to varied chart tasks, explanations, or program traces, as in ChartLlama, ChartAssistant, ChartInstruct, TinyChart, and VProChart [80-90]. For MRAG, this matters because a retriever can index either the visual chart, the table reconstructed from it, or both. In practice, hybrid indexing is usually preferable: tables improve symbolic match and programmatic reasoning, while chart images preserve legends, mark styles, scale choices, color encodings, and formatting that reconstructed tables often omit [77-90,93-101]. [67-101] 8

At the retrieval layer, recent document MRAG systems show a clear movement toward **late-interaction and multivector visual retrieval**. The intellectual lineage runs from dense text retrieval and ColBERT-style late interaction [102,103] through general-purpose contrastive text embeddings [104-106] to document-page retrievers such as ColPali and screenshot-embedding approaches [109,110]. Visual document retrieval methods increasingly represent pages as sets of patch-level or region-level vectors rather than single pooled embeddings, allowing sharper relevance matching for localized evidence such as a caption, table cell, or chart legend [109,128-133]. This trend is visible across ColPali, COLMATE, SERVAL, hybrid-vector retrieval, and region-aware retrieval work [109,128-133,136]. At the same time, MRAG systems rarely rely on a single retrieval pass. M3DocRAG, VDocRAG, VisDoM, MARA, SimpleDoc, ViDoRAG, and LAD-RAG combine multiple retrieval cues—page text, screenshot similarity, reasoning feedback, or modality-specific evidence selection—in order to negotiate the tension between recall and precision [112,120-145]. [102-110,112,120-145] 9

Generation, Reasoning, and Evaluation

Once evidence is retrieved, generation quality depends on **how** the system fuses modalities and controls reasoning. Three families dominate. The first is **evidence concatenation**, in which retrieved screenshots, OCR text, chart tables, or page crops are arranged into a prompt for a multimodal generator. This strategy is simple and often strong, but it scales poorly when evidence count grows or when spatial provenance is needed for every answer [111-124]. The second is **grounded fusion**, where retrieved evidence is aligned with regions or blocks that remain explicit during answer generation. RegionRAG, HKRAG, LAD-RAG, and RECON exemplify efforts to preserve localization and page structure rather than flattening all evidence into text [136,137,140,145]. The third is **reasoning-augmented generation**, where the system interleaves retrieval with program execution, graph traversal, self-reflection, or agentic control. This is especially prominent in chart-centric work—DePlot, MatCha, TinyChart, VProChart, and related methods—because chart QA frequently requires numerical computation, hypothetical comparison, or visual-variable disambiguation [77,78,83,89,90]. It is equally visible in document MRAG systems such as ViDoRAG, HM-RAG, HEAR, and MDocAgent [139-143].

A notable methodological convergence is that chart understanding has become a proving ground for **tool-mediated reasoning within MRAG**. Pure end-to-end generation remains attractive, but chart tasks repeatedly show benefits from intermediate symbolic forms, whether chart tables, executable programs, or mark-level decompositions [73,77-90]. This has broader implications for document MRAG. Financial reports, scientific papers, and slide decks are not merely collections of text-rich images; they contain substructures—tables, figures, plots, forms—whose semantics are often more faithfully manipulated after partial structural extraction. Consequently, the best current systems are increasingly **hybrid** rather than ideological: they use vision-first retrieval when visual fidelity matters, parse-first retrieval when lexical or symbolic operations matter, and tool-assisted reasoning when arithmetic or compositional logic is central [77-90,120-153]. [77-90,120-153] 10

Evaluation has expanded from single-page answer accuracy to a broader assessment of retrieval quality, evidence sufficiency, visual grounding, and robustness. On the document side, DocVQA, InfographicVQA, SlideVQA, DUDE, MMLongBench-Doc, M-LongDoc, LongDocURL, SPIQA, and related datasets measure increasingly difficult combinations of perception and reasoning [52-65]. On the chart side, FigureQA, DVQA, PlotQA, ChartQA, OpenCQA, ChartBench, ChartX, ChartInsights, SCI-CQA, ChartMind, ChartQA-X, and ChartDiff evaluate a spectrum from synthetic low-level perception to open-ended summarization, explanation, and comparative reasoning [67-101]. For MRAG specifically, MRAG-Bench, UDA, MVQA, Dynamic VQA, Chart-MRAG Bench, and document-centric benchmarks such as M3DocRAG, VDocRAG, and VisDoM have begun to separate retrieval from generation and to examine cross-document reasoning [112,115-121,139,150,164]. The key methodological lesson is that answer overlap alone is insufficient. A system can answer correctly for the wrong reasons, can retrieve redundant or non-minimal evidence, or can exploit text-over-visual shortcuts. Strong evaluation therefore requires separate measurement of retrieval recall, grounding quality, citation faithfulness, and robustness under domain or visual perturbation [57-66,86,92-97,115,118,119,150,164]. [57-66,86,92-97,115,118,119,150,164] [11]

Challenges and Prospects

Grounding, Robustness, and Trustworthiness

The first major challenge is evidence faithfulness. In text-only RAG, evidence alignment is already nontrivial; in multimodal settings, it becomes substantially harder because evidence may be visually salient but lexically under-specified, or lexically plausible but visually inconsistent. Several recent systems explicitly target spatial grounding or region selection, yet chart- and document-centric benchmarks continue to reveal modality bias, especially a tendency to over-rely on accompanying text when visual evidence is difficult or noisy [93,115,139,140,145,154-158,164]. This is particularly consequential in charts, where seemingly minor visual attributes—bar color, axis scale, legend assignment, marker shape, or line style—can change the answer. Future work will likely require token- or region-level provenance models whose outputs are auditable, rather than answer-only generation pipelines.

Robustness is a second concern. Poisoning, perturbation, and visual degradation attacks have already been demonstrated for document and MRAG pipelines, including knowledge poisoning, single-image poisoning of visual retrieval, low-level perturbation attacks, and extraction attacks against deployed RAG systems [159-162]. RobustVisRAG further argues that visual degradations are not just a security curiosity but a core evaluation dimension for practical systems [163]. For chart and document MRAG, robustness research must move beyond generic adversarial language attacks and toward **document-native** failure modes: template homogeneity, low-resolution scans, OCR confusion, chart redraw noise, layout shifts, and cross-source inconsistencies between prose and figures. [154-164] [12]

Efficiency, Scalability, and Systems Design

The second major challenge is efficiency. The field increasingly recognizes that visually faithful retrieval is expensive. Multivector vision retrieval improves precision, but it also raises storage, indexing, and latency costs; page-image methods can be more accurate than text-only retrieval, yet they become burdensome for very large corpora [109-113,128-133]. Multi-page and multi-document settings amplify the problem, because candidate evidence proliferates faster than multimodal context windows. Hierarchical retrieval, vector pruning, hybrid single-/multi-vector indexes, and retrieval/reranking cascades are currently the main answers [123-133]. Even so, there is still no settled systems recipe for enterprise-scale repositories with millions of visually similar documents, nor for scientific corpora where one query may require diffuse evidence across many figures and pages.

A related systems issue is the unresolved balance between long-context native MLLMs and retrieval-heavy architectures. Long-context multimodal models have improved substantially, but benchmarks on long documents continue to show diminishing returns from brute-force context expansion alone [57-61]. Retrieval remains attractive because it is selective, modular, and inspectable. Yet retrieval incurs its own error surface: if evidence is missed, even a strong generator fails. The most promising direction appears to be **adaptive systems** that combine retrieval with selective long-context expansion, using coarse retrieval to identify likely evidence neighborhoods and long-context reasoning to integrate them once selected. In this view, long context and retrieval are complements, not substitutes. [57-61,112-145] [13]

Data, Benchmarks, and Domain Generalization

A third challenge concerns data and benchmark design. Many influential datasets remain invaluable, but chart and document understanding still suffers from synthetic biases, template regularities, and evaluation protocols that reward shallow shortcuts [67-76,84-86,92,95-97]. Chart understanding has improved partly because the community moved from synthetic datasets like FigureQA and DVQA toward more realistic settings such as ChartQA, OpenCQA, ChartBench, ChartX, SCI-CQA, and ChartMind [67-76,84-97]. Document MRAG is now undergoing a similar transition from single-page datasets to open-domain, multi-page, and domain-specific benchmarks [56-66,112,120,121,149,150,164]. The next step should be evidence-centered evaluation: datasets should annotate not only answers but also the minimal multimodal evidence set, the required reasoning type, and the acceptable provenance granularity.

Finally, domain generalization remains underdeveloped. Finance, scientific literature, forms, enterprise PDFs, infographics, and multilingual documents each impose distinct priors on layout, terminology, and evidence structure [22,27,32,41,50,62-66,92,149-153]. A retriever that excels on general document QA may struggle on dense scientific figures; a chart model tuned on web charts may fail on flowcharts or low-resource language chart summarization [92,98,99,149-153]. Future progress will likely depend on mixtures of synthetic data generation, domain-specific pretraining, retrieval-aware instruction tuning, and benchmark suites that explicitly test transfer across document genres rather than within a single distribution. [22,27,41,62-66,92,98,99,149-153] [14]

Conclusion

Multimodal retrieval-augmented generation for chart and document understanding has rapidly become a distinct research area rather than a loose overlay of RAG on existing VLMs. Its defining insight is that chart and document intelligence depends on preserving multimodal evidence fidelity while retrieving and reasoning at the right granularity. The most successful systems today are neither purely OCR-based nor purely vision-only; neither purely parametric nor purely retrieval-based; neither purely neural nor purely symbolic. They are hybrid systems that combine document-aware representations, chart-aware structural reasoning, multigranular retrieval, and grounded generation [20-48,74-101,109-153]. The field's immediate frontier is not merely better answer accuracy, but better evidence management: citation-grade grounding, robust multimodal retrieval under real-world distortions, efficient indexing for massive corpora, and benchmarks that measure whether an answer is supported by the right evidence for the right reason. If those goals are met, MRAG will become a general substrate for reliable reading of reports, scientific papers,

financial filings, slide decks, dashboards, and chart-rich knowledge repositories. [9,19,57-66,74-101,109-164]
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